EQ Credit: Low-Emitting Materials

This credit applies to

- BD+C: New Construction (1-3 points)
- BD+C: Core & Shell (1-3 points)
- BD+C: Schools (1-3 points)
- BD+C: Retail (1-3 points)
- BD+C: Data Centers (1-3 points)
- BD+C: Warehouses & Distribution Centers (1-3 points)
- BD+C: Hospitality (1-3 points)
- BD+C: Healthcare (1-3 points)

INTENT

To reduce concentrations of chemical contaminants that can damage air quality and the environment, and to protect the health, productivity, and comfort of installers and building occupants.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Use materials on the building interior that meet the low-emitting criteria below. Points are awarded according to Table 1:

Table 1. Points for low-emitting materials	
2 product categories	1 point
3 product categories	2 points
3 product categories at 90% threshold	3 points
4 product categories	3 points
4 product categories with at least 3 product categories at 90% threshold	3 points + exemplary performance
5 product categories	3 points + exemplary performance

Paints and Coatings

At least 75% of all paints and coatings, **by volume or surface area**, meet the *VOC emissions evaluation* AND 100% meet the *VOC content evaluation*. To meet the 100% requirement for VOC content evaluation, a VOC budget may be used.

The paints and coatings product category includes all interior paints and coatings wet-applied on site, specialized finishes (dyes, sealers, hardeners and toppings for concrete floors), and plaster. Exclude foamed-in place and sprayed insulation (include in Insulation category).

Adhesives and Sealants

At least 75% of all adhesives and sealants, **by volume or surface area**, meet the *VOC emissions evaluation* AND 100% meet the *VOC content evaluation*. To meet the 100% requirement for VOC content evaluation, a VOC budget may be used.

The adhesives and sealants product category includes all interior adhesives and sealants wet-applied on site.

Flooring

At least 90% of all flooring, by cost or surface area, meets the *VOC emissions evaluation OR inherently nonemitting sources criteria, OR salvaged and reused materials criteria*.

The flooring product category includes all types of hard and soft surface flooring (carpet, ceramic, vinyl, rubber, engineered, solid wood, laminates), raised flooring, wall base, transition strips/stair nosing, grills, entryway systems, underlayments, and other floor coverings.

Exclude poured concrete, subflooring (include subflooring in the composite wood category, if applicable),and wet-applied products applied on the floor (include in paints and coatings category).

Wall panels

At least 75% of all wall panels, by cost or surface area, meet the *VOC emissions evaluation, OR inherently nonemitting sources criteria, OR salvaged and reused materials criteria.*

The wall panels product category includes all finish wall treatments (wall coverings, wall paneling, wall tile), gypsum, curtain walls, retail slatwall, trim, interior and exterior doors, non-structural wall framing, interior and exterior windows, window film and treatments, countertops, laminate/veneer used for built-in cabinetry, non-structural sandwich panels, CMU.

Exclude cabinetry (include the composite wood components of built-in cabinetry in the composite wood category and free-standing cabinetry in the furniture category), and vertical structural elements (include structural wood panels or structural composite wood in the composite wood category, if applicable), bathroom accessories, and door hardware.

Ceilings

At least 90% of all ceilings, by cost or surface area, meet the *VOC emissions evaluation, OR inherently nonemitting sources criteria, OR salvaged and reused materials criteria.*

The ceilings product category includes all ceiling panels, ceiling tile, surface ceiling structures such as gypsum or plaster, suspended systems (including canopies and clouds), and glazed skylights.

Exclude overhead structural elements (include structural elements in the composite wood category, if applicable).

Insulation

At least 75% of all insulation, by cost or surface area, meets the *VOC emissions evaluation.*

The insulation product category includes all thermal and acoustic boards, batts, rolls, blankets, sound attenuation fire blankets, foamed-in place, loose-fill, blown, and sprayed insulation.

Exclude insulation for HVAC ducts and plumbing piping from the credit. Insulation for HVAC ducts may be included at the project team's discretion.

Furniture

At least 75% of all furniture in the project scope of work, by cost, meets the *furniture emissions evaluation, OR inherently nonemitting sources criteria, OR salvaged and reused materials criteria.*

The furniture product category includes all seating, desks and tables, filing/storage, free-standing cabinetry, systems furniture, moveable/demountable partitions, bathroom/toilet partitions, shelving, lockers, specialty and custom fixtures and furniture, and furnishing items (such as area rugs, cubicle curtains, mattresses, and mirrors) purchased for the project.

Exclude office and bathroom accessories, art, recreational items, and planters from the credit.

Composite Wood

At least 75% of all composite wood**, by cost or surface area**, meets the *Formaldehyde emissions evaluation OR salvaged and reused materials criteria.*

The composite wood product category includes all particleboard, medium density fiberboard (both medium density and thin), hardwood plywood with veneer, composite or combination core, and wood structural panels or structural wood products.

Exclude products covered in the flooring, ceiling, wall panels, or furniture categories from this category.

Low-emitting criteria

Inherently nonemitting sources

Product is an inherently nonemitting source of VOCs (stone, ceramic, powder-coated metals, plated or anodized metal, glass, concrete, clay brick, and unfinished or untreated solid wood) and has no binders, surface coatings, or sealants that include organic chemicals.

Salvaged and reused materials

Product is more than one year old at the time of use. If finishes are applied to the product on-site, the finishes must meet the *VOC emissions evaluation* AND *VOC content evaluation* requirements.

VOC emissions evaluation

Product has been tested according to California Department of Public Health (CDPH) Standard Method v1.2–2017 and complies with the VOC limits in Table 4-1 of the method. Additionally, the range of total VOCs after 14 days (336 hours) was measured as specified in the CDPH Standard Method v1.2 and is reported (TVOC ranges: 0.5 mg/m^3 or less, between 0.5 and 5 mg/m^3, or 5 mg/m^3 or more).

Laboratories that conduct the tests must be accredited under ISO/IEC 17025 for the test methods they use. Products used in any setting other than schools and classrooms must be modeled to private office scenario. For schools projects, modeling to office and/or schools scenario is permitted.

The statement of product compliance must include the exposure scenario(s) used, the range of total VOCs, and must follow the product declaration guidelines in CDPH Standard Method v1.2-2017, Section 8. Manufacturer statements must also include a summary report from the laboratory that is less than three years old and the amount of wet-applied product applied in mass per surface area (if applicable). Organizations that certify manufacturers' claims must be accredited under ISO/IEC 17065.

VOC content evaluation

Product meets the VOC content limits outlined in one of the applicable standards and for projects in North America, methylene chloride and perchloroethylene may not be intentionally added.

Statement of product compliance must be made by the manufacturer or a USGBC-approved third-party. Any testing must follow the test method specified in the applicable regulation. If the applicable regulation requires subtraction of exempt compounds, any content of intentionally added exempt compounds larger than 1% weight by mass (total exempt compounds) must be disclosed.

- Paints and coatings:
    - California Air Resource Board (CARB) 2007 Suggested Control Measure (SCM) for Architectural Coatings

    - South Coast Air Quality Management District (SCAQMD) Rule 1113, amended February 5, 2016, effective date 1/1/19
- Adhesives and sealants:
    - SCAQMD Rule 1168, amended October 6, 2017

Formaldehyde emissions evaluation

Product meets one of the following:

- Certified as ultra-low-emitting formaldehyde (ULEF) product under EPA Toxic Substances Control Act, Formaldehyde Emission Standards for Composite Wood Products (TSCA, Title VI) (EPA TSCA Title VI) or California Air Resources Board (CARB) Airborne Toxic Control Measure (ATCM)
- Certified as no added formaldehyde resins (NAF) product under EPA TSCA Title VI or CARB ATCM
- Wood structural panel manufactured according to PS 1-09 or PS 2-10 (or one of the standards considered by CARB to be equivalent to PS 1 or PS 2) and labeled bond classification Exposure 1 or Exterior
- Structural wood product manufactured according to ASTM D 5456 (for structural composite lumber), ANSI A190.1 (for glued laminated timber), ASTM D 5055 (for I-joists), ANSI PRG 320 (for cross-laminated timber), or PS 20-15 (for finger-jointed lumber).

Furniture emissions evaluation

Product has been tested in accordance with ANSI/BIFMA Standard Method M7.1–2011 (R2016) and complies with ANSI/BIFMA e3-2014e or e3-2019e Furniture Sustainability Standard, Sections 7.6.1 (for half credit, by cost) OR 7.6.2 (for full credit, by cost), OR 7.6.2 AND 7.6.3 for one and a quarter credit, by cost. Laboratories that conduct the tests must be accredited under ISO/IEC 17025 for the test methods they use.

Seating products must be evaluated using the seating scenario. Classroom furniture must be evaluated using the standard school classroom scenario. Other products should be evaluated using the open plan or private office scenario, as appropriate. The open plan scenario is more stringent.

Statements of product compliance must include the exposure scenario(s). Organizations that certify manufacturers' claims must be accredited under ISO/IEC 17065.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The compliance methodology for this credit is more straightforward but continues to promote holistic consideration of products installed in the building and their potential overall impact on indoor air quality

Option 2 Budget Calculation method has been removed completely to simplify the approach towards compliance requirements and core credit achievement is now based solely on meeting number of compliant product categories. Previously bundled product category of ceilings, walls and insulation available as separate product categories, updated definitions are added for all product categories and compliance criteria for most product categories can now be showed either via percent of cost or surface area for most product categories. The threshold ranges for compliance are adjusted to 75%-90% by cost or surface area. Some other changes are: i) including inherently non-emitting sources and salvaged/reused materials as part of compliance criteria rather than exceptions/exclusions to reward project teams more directly, (ii) clear allowance for using VOC budget approach for VOC content for wet-applied products, and iii) modifying existing standard references CDPH standard method v1.1-> CDPH standard method v1.2 (updated version).

Other standards further modified were as follows:

- SCAQMD Rule 1113, June 2011 → SCAQMD Rule 1113, February 2016 (updated version)
- SCAQMD Rule 1168, July 2005 → SCAQMD Rule 1168, October 2017 (updated version)
- ANSI/BIFMA M7.1-2011 → ANSI/BIFMA M7.1-2011 (R2016) (re-affirmed version)
- ANSI/BIFMA e3-2011 → ANSI/BIFMA e3-2014e (updated version) and ANSI/BIFMA e3-2019 (updated version) EPA TSCA Title VI and structural composite wood industry standards (new standard for formaldehyde emissions in composite wood)

Additionally, standards for projects outside of the U.S. are now available for use through regional compliance pathways (see International Tips section).

Step-by-Step Guidance

Step 1. Research products

Review project documents to identify all applicable products and specify them according to the low-emitting criteria appropriate for the product.

In most cases, turning to a third-party certification program is the easiest way to find and specify products.

- VOC emissions evaluation: See the list of certification programs that use CDPH Standard method v1.2 published by the California Department of Public Health at cdph.ca.gov. Green Seal GS-11 Edition 4.0 and Asthma & allergy friendly Certification Standard (ASP:04-01 Paint Indoor Decorative Wall, ASP: 05-01 Resilient Flooring, ASP:05-03 Textile Flooring, and ASP:19-01 Fiberglass Insulation) also use CDPH Standard method v1.2.

- VOC content evaluation: The following programs certify VOC content to CARB SCM, SCAQMD Rule 1113 and/or SCAQMD Rule 1168:
    - Asthma & allergy friendly Certification Standard (ASP:04-01 Paint Indoor Decorative Wall)
    - Certified Green by MAS
    - Green Seal GS-11 Edition 4.0
    - Indoor Advantage Gold – Building Materials by SCS
    - ClearChem by Berkeley Analytical (*self-declared claims*)

- Formaldehyde emissions evaluation:
    - See the list of third-party certifiers for TSCA Title VI and CARB ATCM published by the EPA at epa.gov.
    - Structural composite wood products with an APA trademark meet the requirement of being PS 1-09 or PS 2-10 and labeled bond classification Exposure 1 or Exterior.

- Furniture emissions evaluation: The following certifications and programs use ANSI/BIFMA e3 and are accredited under ISO/IEC 17065:
    - *Clean Air Silver by Intertek (for half credit)*
    - Clean Air Gold by Intertek
    - *GREENGUARD Certified by UL (for half credit)*
    - GREENGUARD Gold by UL
    - *Indoor Advantage-Furniture by SCS (for half credit)*
    - Indoor Advantage Gold- Furniture by SCS
    - Level by BIFMA
    - Certified Green by MAS
    - VOC Green by Benchmark International

For the VOC content evaluation, when determining the appropriate VOC limit, use the manufacturer's product classification rather than how the product was used on the project. For example, a defined roof coating is not a carpet adhesive simply because it was used this way on the project. If the product is classified as a roof coating under the stated regulation (such as SCAQMD), it must meet the appropriate VOC limit for roof coatings.

Follow the language in the rating system for each product category to identify materials to include. Some materials installed in the project are excluded from this credit entirely. For example, the following products are not applicable to the product categories, or they are considered exterior products: equipment related to fire suppression, HVAC, plumbing, electrical, conveying and communications systems, poured concrete, Structural insulated panels (SIPs), and water-resistive barriers (material installed on a substrate to prevent bulk water intrusion). For this credit, the following products should be considered within the building interior: the air barrier membrane itself, the vapor barrier/vapor retarder membrane (if used inside the air barrier), and non-structural prefabricated building envelope products that are in contact with the building interior such as sandwich panels,

Step 2. Install compliant products

Make sure selected products are installed in the building. Any product substitutions should be carefully reviewed by the design team and contractor for compliance with credit requirements. At time of product purchase, collect documentation that confirms the product complies with the low-emitting criteria.

Step 3. Calculate percentage compliance

For each product category being pursued, list out the material, cost/surface area/volume, and compliant criteria. Confirm the percentage compliant meets the threshold for the given product category.

For VOC content, note that the 100% compliance threshold may be met using a VOC budget approach. Refer to the LEED v4 reference guide Option 1. Step 1 for more information on this calculation.

For the Furniture product category, the contributing cost is dependent on the compliance criteria selected. Make sure the actual product cost AND contributing product cost are listed in the calculations.

Further Explanation

Composite Wood

TSCA Title VI (Formaldehyde standards for Composite Wood Products Act): In 2016, EPA issued a final rule to implement the formaldehyde standards for composite wood products act that added Title VI to the Toxic Substances Control Act (TSCA). TSCA Title VI establishes formaldehyde emission standards identical to the California Air Resources Board (CARB) limits. As with CARB ATCM, the goal is to reduce exposure and adverse effects from formaldehyde emissions in composite wood. The rule affects formaldehyde emission standards applicable to hardwood plywood (with some exemptions for laminated products under hardwood plywood), medium density fiberboard and particleboard and finished goods containing these products that are manufactured and traded in the U.S. region., establishes a third party certification program for emission testing of these products and includes requirements for ULEF and NAF resins used in these products.

If the composite wood product has a finish or treatment applied off-site (including but not limited to: fire-retardant, paint, stain or other coating), the product may still be included in the composite wood category if the off-site applied finish or treatment complies with the VOC emissions evaluation. To document this situation, include the VOC emissions evaluation documentation as supporting documentation for the composite wood product.

Wet-applied products, applied to composite wood on-site, belong in the adhesives and sealants or paints and coatings category, as applicable.

Documentation for Manufacturer Claims for VOC emissions evaluation

If compliance to CDPH or EN 16516 is being demonstrated through a manufacturer's claim, provide a test report or summary report with the following information:

CDPH Standard Method v1.2-2017
Declaration that the product has been tested according to CDPH Standard Method v1.2-2017 and complies with the VOC limits in Table 4-1 of the method;
TVOC results at 14 days measured as specified in CDPH Standard Method v1.2-2017;
Test date
The name of the laboratory that performed the evaluation and documentation (such as accreditation number or certificate with scope of accreditation) demonstrating the accreditation under ISO/IEC 17025 for the test method
The modeling scenario used (must be private office unless the product is installed in a classroom)
For wet-applied products, the amount of product applied in mass per surface area (during testing)

EN 16516:2017 with 2018 updates
Declaration that the product has been tested according to EN 16516:2017 with 2018 updates and complies with the German AgBB Testing and Evaluation Scheme (2018) thresholds.
Sum of R ratio results for VOCs with LCI values and summed individual VOC concentrations for VOCs without LCIs.
Declaration that formaldehyde limit of 10 µg/m3 after 28 days has been met

TVOC value measured after 28 days and is < 1000 ug/m3
Test date
The name of the laboratory that performed the evaluation and documentation (such as accreditation number or certificate with scope of accreditation) demonstrating the accreditation under ISO/IEC 17025 for the test method

Inherently Nonemitting Materials

Stone, ceramic, and porcelain tiles; powder-coated metals, plated or anodized metal, glass, clay brick, and solid wood are materials which, on their own, are considered inherently nonemitting without additional information.

For any other materials, a manufacturer chemical inventory of the product to at least 0.1% (1000 ppm) *(such as those used to document MRc Material Ingredients*) is required to confirm the product complies with the inherently nonemitting sources criteria.

- Consider the entire product and the fabrication. For example, a hollow metal door is made of other materials in addition to metal, a window is comprised of more than glass, and concrete often includes more than stone, gravel, sand, cement and water (e.g. additives like fly ash, plasticizers and hardeners).
- When an inherently nonemitting material is joined with binders/resins, other materials (ex. gaskets, core/internal components); has a surface coating, treatment, or adhesives/sealants, etc.) the resultant product is subject to the VOC emissions evaluation criteria.
- A wet-applied product with 0 g/L of VOC content is not considered inherently nonemitting.

International standards

See *International tips* section below

International Tips

Projects outside of the U.S. may want to consider these additional certifications and programs that have been identified to meet the low-emitting materials credit requirements. *Note: This list is not comprehensive. Additional programs may be used if they meet the applicable rating system requirements or regional compliance pathways.*

- VOC emissions evaluation: The following certifications and programs use EN 16516 and may be used to comply with the regional ACP for VOC emissions evaluation:
    - *Blue Angel (if formaldehyde limit of 10 micrograms per cubic meter after 28 days is also met)*
    - *Danish indoor climate label Emission Class 1*
    - *Eco-INSTITUT-Label (if formaldehyde limit of 10 micrograms per cubic meter after 28 days is also met)*
    - *EMICODE EC1 (if formaldehyde limit of 10 micrograms per cubic meter after 28 days is also met. EMICODE has formaldehyde limit of 50 $\mu g/m^3$ after 3 days)*
    - *EMICODE EC1^{PLUS} (Additional information regarding the formaldehyde limit at 28 days is not required for products meeting EMICODE EC1plus)*
    - Finnish Emission Classification of Building Materials (M1)
    - GUT (for textile floorings)
    - Indoor Air Comfort Gold
    - *NaturePLUS (if formaldehyde limit of 10 micrograms per cubic meter after 28 days is also met.)*
    - *TÜV SÜD guidelines TM-07, TM-09, and TM-10 (if formaldehyde limit of 10 micrograms per cubic meter after 28 days is also met.)*

- VOC emissions evaluation: The following certification uses CDPH Standard Method v1.2-2017 and may be used by projects outside of the U.S. to comply with VOC emissions evaluation.
    - Taiwan Healthy Building Material Label
- VOC content evaluation: The following certifications and programs use European Decopaint Directive and TRGS 610 and may be used to comply with the regional ACP for VOC content evaluation:
    - Indoor Air Comfort Gold
    - EMICODE EC1 or EC1 Plus
- Furniture emissions evaluation
    - Finnish Emission Classification of Building Materials (M1)
    - TÜVRheinland Green Product Mark Furniture
- Formaldehyde emissions evaluation (for composite wood)
    - Blue Angel Composite Wood Panels
    - Byggvarubedömningen (BVB)
    - Finnish Emission Classification of Building Materials (M1)
    - French VOC Emissions Labeling Class A or A+
    - Indoor Air Comfort
    - Indoor Air Comfort Gold

Required Documentation

- USGBC's Low-emitting materials calculator or equivalent documentation
- Documentation of low-emitting criteria met
    - Inherently nonemitting: chemical inventory of the product to at least 0.1% (1000 ppm) confirming product is made from inherently nonemitting materials (*not required if product is stone, ceramic, and porcelain tiles; powder-coated metals, plated or anodized metal, glass, clay brick, and solid wood)*
    - Salvaged and reused: applicable purchase order, finish plan, schedule, and/or specifications from contract documents
    - VOC emissions evaluation:
        - Third-party claim: product certificate from third party certifier with modeling scenario
        - Manufacturer claim (also called self-declared or first-party claims)
            - Laboratory test report or summary report from the manufacturer or the laboratory that includes the required information outlined in Further Explanation: Documentation for Manufacturer Claims for VOC emissions evaluation
    - VOC content: product certificate from third party certifier or manufacturer information documenting the VOC content in grams per liter (g/L) less water and federally exempt solvents.
    - Formaldehyde emissions: one of the following:
        - Product certificate from third-party certifier that states the certification period
        - Product information or manufacturer statement of conformity
        - Photograph of product's APA trademark stamp (for structural products)
        - The VOC emissions evaluation documentation for any finish or treatment applied off-site (if applicable).
    - Furniture emissions: product certificate from third-party certifier that includes the exposure scenario used. Also include the BIFMA LEVEL® scorecard (if the BIFMA Level® e3 Sections achieved are not included on the third party certificate).

Exemplary Performance

Earn 5 product categories or earn 4 product categories and reach 90% threshold in at least three product categories.

Connection to Ongoing Performance

- LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to reduce chemical contaminant levels for improved air quality and human health such as using inherently non-emitting products and/or using products with low VOC content/emissions in newly constructed spaces can contribute to better indoor environmental quality during operations phase.